COMP3013 DBMS Lab 5

United International College

Aggregation

- Suppose that we want to count the number of different languages.
- The aggregation function **COUNT** can be used.

```
SELECT COUNT(*) FROM language
```

- "Aggregation" means performing some operations on a **group of objects**.
- In this example, all tuples in the table "language" is considered as a group. The query counts the number of distinct objects in the group.
- After applying an aggregation function, the result of **COUNT(*)** is treated as an attribute.

GROUP BY

- For example, find the number of films for each language. The resulting table has two columns, one for languages, one for the number of films.

- Only those films of the same language are in the same group (different language different group).
- Then, count the number within each group.
- To achieve the grouping, we use **GROUP BY**.

```
SELECT language_id, count(film_id)
FROM film JOIN language USING(language_id)
GROUP BY (language_id)
```

- To indicate the result of aggregation of each group, group by attributes are usually selected.

GROUP BY

- The GROUP BY clause contains a set of attributes.
- Tuples with the same value of **all** group by attribute(s) are in the same group.
- Be careful when the group by attribute is not a key.
- Compare the two queries and tell the difference.

```
SELECT first_name, COUNT(film_id)
FROM actor JOIN film_actor USING(actor_id)
GROUP BY first_name

SELECT actor_id, first_name, COUNT(film_id)
FROM actor JOIN film_actor USING(actor_id)
GROUP BY actor_id
```

HAVING

- Sometimes we need condition checking before and after aggregation functions.
- For example, show the actors' names and the number of Sci-Fi movies played by him/her if the number of Sci-Fi movies is more than 3.

```
SELECT actor_id,first_name,last_name,COUNT(film_id)
FROM actor JOIN film_actor USING (actor_id)
      JOIN film USING(film_id)
      JOIN film_category USING(film_id)
      JOIN category USING(category_id)
WHERE category.name='sci-fi'
GROUP BY actor_id
HAVING COUNT(film_id)>3;
```

The query is executed in this sequence.

1. Combine the tables in `FROM` clause
2. Check the condition in `WHERE` clause
3. Group tuples
4. Aggregation function
5. Check the condition in HAVING clause
6. Output the selected attributes

Aggregation

- Other than `COUNT`, there are more aggregate functions for different purposes.
  - **`MAX`**: find the maximum value
  - **`MIN`**: find the minimum value
  - **`AVG`**: calculate the average value
  - **`SUM`**: sum up the values

Example

- Find the number of films in each category.
- Count the money spent on film rental by Elizabeth Brown.
- Split the money spent by Elizabeth Brown into different categories.

Example

- Find the number of films in each category.

```
SELECT name,count(film_id)
FROM category JOIN film_category USING (category_id)
GROUP BY category_id
```

Example

- Count the money spent on film rental by Elizabeth Brown.

```
SELECT SUM(amount)
FROM customer JOIN payment USING(customer_id)
WHERE first_name='Elizabeth' AND last_name='Brown'
```

Example

- Split the money spent by Elizabeth Brown into different categories.

```
SELECT category.name, SUM(amount)
FROM customer JOIN payment USING(customer_id)
     JOIN rental USING(rental_id)
     JOIN inventory USING(inventory_id)
     JOIN film_category USING(film_id)
     JOIN category USING(category_id)
WHERE first_name='Elizabeth' AND last_name='Brown'
GROUP BY category_id
```

Distinct

- Assuming that every country has a city. Why the following query cannot find the number of countries?

```
SELECT COUNT(country_id)
FROM city
```

Distinct

- We only assume that every country has a city. But, a country may have multiple cites.
- Those countries will be over counted.
- We can use **`DISTINCT`** to remove duplications.

```
SELECT COUNT(DISTINCT country_id)
FROM city
```

Exercises

Write SQLs for the following questions.

1. Calculate the average rental duration among all films.
2. Calculate the number of films rented by Norman Currier.
3. Calculate the number of films rented by each customer.
4. Calculate the number of films rented by each group of customers from the same country.
5. Calculate the number of horror films rented by each group of customers from the same country.
6. Find the countries which have some customers who have rented more than 4 horror films.

Suppose a film is only counted once if it is rented multiple times. Be careful with the duplications.

Save your queries in a txt file. Rename it as “COMP3013 Lab5 ###.txt”, where “###” is your student ID. And submit it on iSpace. The DDL is 24 hours after the lab.

End of Lab 5